Neutral-Atom Quantum Computing Architecture for Cryptographic Simulation

Neutral-atom quantum processors exploit optical tweezer arrays to trap individual rubidium and ytterbium atoms within ultra-high vacuum chambers. Two-qubit gate operations are mediated via laser excitation to high-lying Rydberg states, providing strong, switchable dipole-dipole interactions with high physical fidelity.

Our research constructs an error-mitigated quantum simulation circuit to benchmark surface code lattice thresholds against coherent laser phase noise and spontaneous atomic emission events.